

MDR SENTINEL: HOSPITAL OUTBREAK INTELLIGENCE

24GE411-PRODUCT DEVELOPMENT LAB-2 REPORT

Submitted by

RAVIPROLU LAKSHMI SRI DIVYA

111520102180

REVATHI V

111520102181

PUTHA DHATRIKA

111520102171

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



R.M.D. ENGINEERING COLLEGE
(An Autonomous Institution)
KAVARAIPETTAI-601 206

MARCH 2026

MDR SENTINEL: HOSPITAL OUTBREAK INTELLIGENCE

24GE411-PRODUCT DEVELOPMENT LAB-2 REPORT

Submitted by

RAVIPROLU LAKSHMI SRI DIVYA **111520102180**

REVATHI V **111520102181**

PUTHA DHATRIKA **111520102171**

in partial fulfillment for the award of the degree

of

BACHELOR OF

ENGINEERING IN

COMPUTER SCIENCE AND ENGINEERING



R.M.D. ENGINEERING COLLEGE
(An Autonomous Institution)
KAVARAIPETTAI-601 206

MARCH 2026

BONAFIDE CERTIFICATE

Certified that this project report “**MDR SENTINEL HOSPITAL OUTBREAK INTELLIGENCE**” is the Bonafide work of “**RAVIPROLU LAKSHMI SRI DIVYA (111524102180), REVATHI V (111524102181), PUTHA DHATRIKA (11152102171)**” who carried out the project work under my supervision.

SIGNATURE

Dr. P EZUHMALAI

B.Tech, M.Tech, FIE, Ph.D

HEAD OF THE DEPARTMENT,

Professor,

Computer Science and
Engineering,

R.M.D. Engineering College.

RSM Nagar, Kavaraipeṭṭai.

SIGNATURE

Dr. V SHARMILA

B.E, M.E., Ph.D

Assistant Professor

Computer Science and Engineering,

R.M.D. Engineering College,

RSM Nagar, Kavaraipeṭṭai.

VIVA – VOCE EXAMINATION

The Viva – Voce Examination of the student who submitted this Product Development work held on

.....

RAVIPROLU LAKSHMI SRI DIVYA

(111524102180)

REVATHI V

(111524102181)

PUTHA DHATRIKA

(111524102171)

INTERNAL EXAMINER

EXTERNAL EXAMINER

DEPARTMENT VISION

To produce intellectual software professionals by empowering them to compete globally towards professional excellence

DEPARTMENT MISSION

- To impart modern teaching methodologies to provide quality technical education to the students
- To establish a work environment to practice new software tools and languages
- To identify industrial requirements and produce employable engineers
- To engage students and staff members in research to meet the requirements of software industry

PROGRAMME EDUCATIONAL OBJECTIVES

PEO 1

To enable graduates to pursue higher education and research, or have a successful career in industries associated with Computer Science and Engineering, or as entrepreneurs.

PEO 2

To ensure that graduates will have the ability and attitude to adapt to emerging technological changes.

PEO 3

To enable graduates to value social responsibility and possess strong leadership and interpersonal abilities.

PROGRAMME SPECIFIC OUTCOMES (PSOs)

PSO 1

To analyze, design and develop computing solutions by applying foundational concepts of Computer Science and Engineering.

PSO 2

To apply software engineering principles and practices for developing quality software for scientific and business applications.

PSO 3

To adapt to emerging Information and Communication Technologies (ICT) to innovate ideas and solutions to existing/novel problems.

PROGRAM OUTCOMES (POS)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

Course Outcomes:

CO1	Identify real-world problems and formulate appropriate problem statements for product development.
CO2	Analyze user needs and apply design thinking principles to develop innovative product concepts.
CO3	Develop functional prototypes using suitable hardware/software tools and technologies.
CO4	Integrate multidisciplinary knowledge to design, test, and validate product performance.
CO5	Demonstrate teamwork, project management, and communication skills through effective documentation and presentation of the developed product.
CO6	Evaluate the social, ethical, and environmental impact of the developed product and propose improvements for sustainable innovation.

CO / PO Mapping:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO1	3	3	2								3	2	2	1
CO2	2	3	2	2		3					3	2	2	1
CO3	1	1	3	3	3				1		3	1	1	2
CO4	1	1	2	3				2	3		3	1	1	2
CO5	1	1		1		3		3	3	2	3	1	1	2
CO6	2	1	1	1		2	3	1		1	3	1	1	1

Low-1, Medium-2, Strong-3

ACKNOWLEDGEMENT

We take this opportunity to put to record my sincere thanks to all who enlightened our path towards the successful completion of this project. At the very outset, we thank the almighty for his abundant blessing on us.

It is our greatest privilege to convey our thanks to Chairman **SHRI. R.S. MUNIRATHINAM** and Vice-Chairman **SHRI R.M. KISHORE B.E., MBA.**, for having provided us with all required facilities to complete our project without hurdles.

We take this opportunity to give profound and heartfelt thanks to the Principal **Dr.N.ANBUCHEZHIAN B.E.,M.S.,MB.A.,M.E.,Ph.D.**, for his continuous support in successful completion of this project.

It is our greatest privilege to convey our thanks to the Dean Research **Dr. K. K. THYAGHARAJAN B.E., M.E., Ph.D.**, for having provided us with all required facilities to complete our project work without hurdles.

We express our sincere feeling of gratitude to our Head of the Department of Computer Science and Engineering **Dr. P.EZHUMALAI B.E., M.Tech, Ph.D.**, for his advice and encouragement in all stages of my project work.

It is our pleasant duty to deliver hearty thanks to the extraordinary efforts of our Product Development Lab coordinator Department of Computer Science and Engineering, **Dr. V. SHARMILA, B.E, M.E., Ph.D.**, for her continuous guidance and suggestions, without her this project would not have become a reality.

We owe our sincere thanks to all the faculty of the Department of Computer Science and Engineering for their constant support and encouragement towards the completion of Product Development Lab

ABSTRACT

Multi-Drug Resistant (MDR) infections are a significant problem for contemporary healthcare infrastructures, causing prolonged hospitalization, increased healthcare costs, and increased mortality rates. Current methods for infection control involve traditional techniques such as manual contact tracing, disjointed data management, and delayed laboratory reporting, which are more reactive in nature rather than preventive in nature. This report outlines an intelligent system for infection surveillance and outbreak prevention using AI technology, which would enable real-time surveillance and control of MDR infections in hospital settings. The proposed system would incorporate Bluetooth Low Energy technology for indoor patient and staff movement tracking, Internet of Things technology for real-time sensing and data collection, and digital twin technology for simulating real-time patient and staff movement in hospital settings. Advanced graph analysis techniques would be used for detecting concealed infection chains and predicting super-spreader events, and machine learning techniques would be used for predicting infection risks in real-time. The proposed system would provide a comprehensive platform for consolidating Electronic Health Records, laboratory reports, and sensor data for real-time infection control and decision support. Furthermore, real-time alerts and infection simulation using Infection Replay Engine would provide real-time response and root cause analysis for infection control. The proposed system would be more in tune with contemporary healthcare infrastructure requirements and would contribute to building resilient hospitals equipped with technology and AI-based systems for patient care and safety.

TABLE OF CONTENTS

CHAPTER NUMBER	TITLE	PAGE NUMBER
	ACKNOWLEDGEMENT	vii
	ABSTRACT	viii
I	INTRODUCTION	1
	SCOPE OF PROJECT OBJECTIVE PROBLEM STATEMENT	
II	LITERATURE SURVEY	4
III	METHODOLOGY	8
	EXISTING SYSTEM PROPOSED SYSTEM SYSTEM ARCHITECTURE APPLICATIONS SYSTEM REQUIREMENTS	
IV	RESULTS	18
V	CONCLUSION	23
VI	REFERENCES	24

LIST OF FIGURES

FIGURE NUMBER	TITLE	PAGE NUMBER
3.3.1	SYSTEM ARCHITECTURE	10
3.5.1	ER DIAGRAM	13
3.5.2	CLASS DIAGRAM	16
4.1	DASHBOARD PAGE	18
4.2	ICT DASHBOARD	19
4.3	ICT DASHBOARD (WHAT IF SIMULATOR)	20
4.4	DOCTOR DASHBOARD	21
4.5	NURSE DASHBOARD	22

CHAPTER-1

INTRODUCTION

Multi-Drug Resistant (MDR) infections have presented a critical problem in contemporary healthcare, resulting in extended hospital stays, increased healthcare expenses, and escalating mortality rates. The majority of hospitals rely on traditional and time-consuming contact tracing, delayed test results, and fragmented data systems, which makes infection control slow and passive.

The project suggests an AI-powered infection intelligence system, which will allow hospitals to monitor and control infections in real time and prevent outbreaks. The system will utilize technology such as Bluetooth Low Energy, Internet of Things, and digital twin modelling to monitor the movements and interactions of patients and healthcare staff within the healthcare setting.

The system will utilize Electronic Health Records, test results, and sensor data to provide a unified infection control system. The system will utilize graph-based analytics and machine learning to identify exposure patterns, high-risk contacts, and potential outbreaks. The system will include features such as infection alerts and infection replays, which will enhance effective decision-making. The system will revolutionize traditional infection control by providing a data-driven approach, which will enhance efficiency and safety.

SCOPE OF THE PROJECT:

The main aim of the project is to develop an intelligent system that can assist the hospital in the detection, monitoring, and prevention of the spread of MDR infections. The project utilizes the latest technology, including Artificial Intelligence, IoT, and Bluetooth Low Energy. It includes the integration of data from sensors and hospital records.

The system represents the hospital environment as a digital model. It tracks the movement of patients as well as hospital staff. It stores data, including the location of the patients, the duration of the contact, and the records of the patients. It helps the hospital authorities easily monitor the spread of infections.

OBJECTIVES:

The aim of the project is to develop a smart system of infection control within a short period of time. It utilizes the integration of multiple technologies. It ensures the real-time monitoring, prediction, and management of the spread of infections. The main aim of the project is the interconnection of hospital data, the movement of the patients, as well as the sensors.

Key Objectives:

- It provides effective real-time monitoring of infections.
- It ensures the effectiveness of contact tracing.
- It provides the ability to identify potential infections.
- It ensures the identification of high-risk areas.
- It provides the ability to integrate data from electronic health records, lab, as well as sensors.
- It ensures the effectiveness of the alert system.
- It provides the ability to analyse the infections.
- It ensures the effectiveness of the system of infection control.
- It provides the ability to enhance the safety of the patients.
- It ensures the effectiveness of the system.

PROBLEM STATEMENT:

Modern healthcare infrastructures are in a critical situation due to the increasing rate of Multi-Drug Resistant (MDR) infections. These infections are significantly impacting patient safety, treatment costs, and increasing hospital stays. In spite of advances in medical technologies, most hospitals are still relying on traditional infection control practices such as tracing contacts from records, delayed laboratory reporting, and segregated data management. It is very challenging to track the spread of infections using traditional methods such as tracing contacts from records, delayed laboratory reporting, and segregated data management.

Without a proper system in place, hospitals are unable to utilize data from Electronic Health Records, laboratory reports, and real-time patient movement. In most cases, hospitals are unable to track the spread of infections in a timely manner. Infection outbreaks are identified very late in the process when they have already started to spread among patients, healthcare staff, and in the environments of hospitals. In such a situation, the efficiency of such traditional methods is very poor in controlling the spread of infections.

To resolve such difficulties, there is a need to develop a smart solution based on technology to monitor, track, and predict the spread of infections in hospitals. In such a solution, better coordination, more efficient response, and decision-making can be achieved to improve healthcare safety.

In such a context, the following specific problems are identified to be addressed in this project:

- Who are the users of such a system?
- What technologies are to be used for such a solution?
- How can a real-time digital monitoring system help in reducing the spread of MDR infections?
- How can a real-time digital monitoring system help in predicting and preventing infections?
- Can a significant difference in infection control be achieved between traditional and smart infection control?

CHAPTER-II

LITERATURE SURVEY

PAPER-1

TITLE: Study of Hospital Emergency Resource Scheduling Based on Digital Twin Technology

AUTHOR: Xiaohua Hu; Hongwei Cao; Jia Shi; Yonghui Dai; Weihui Dai

YEAR: 2022

INTRODUCTION:

Hospital emergency management plays a critical role in ensuring the effective functioning of healthcare systems, especially during public health crises. The importance of efficient emergency management has significantly increased in recent years, particularly after the global outbreak of COVID-19. Hospitals are required to respond quickly and efficiently to sudden surges in patient numbers, while simultaneously managing limited resources such as medical personnel, protective equipment, and essential supplies. This makes emergency management a highly complex and dynamic process. One of the major challenges in hospital emergency management is the optimal allocation and scheduling of resources. These resources include healthcare workers, medical equipment, and essential supplies such as masks, gloves, protective clothing, and testing kits. During emergencies, the demand for these resources fluctuates rapidly, requiring hospitals to prioritize critical departments such as fever clinics and isolation wards. Improper allocation can lead to resource shortages, reduced efficiency, and compromised patient care. Emergency resource scheduling is therefore a key component of hospital management. However, it is often affected by uncertainty and variability in demand, making decision-making difficult. To address these challenges, advanced technologies such as digital twin systems and virtual reality interaction are being explored. Digital twin technology enables real-time monitoring and simulation of hospital operations, while virtual reality enhances decision-making through interactive visualization. This study focuses on improving the perception and allocation of emergency resources using these advanced technologies. By developing models for data integration, medicine recognition, and dispatch mechanisms, it aims to provide a more efficient and reliable approach to hospital emergency management.

LINK: [Study of hospital emergency resource scheduling based on digital twin technology | IEEE Conference Publication | IEEE Xplore](#)

PAPER-2

TITLE: Parallel Hospital: Acp-Based Hospital Smart Operating System

AUTHORS: Yongjun Wang; Fei-Yue Wang; Xiao Wang; Zhongjun Guan; Liwei

Ouyang; Jiangong Wang

YEAR: 2021

INTRODUCTION:

With the rapid advancement of intelligent technologies, healthcare systems are evolving toward more efficient and adaptive models of management. One such innovative concept is the parallel hospital, which is based on the principles of parallel medicine. The idea of a parallel hospital emphasizes the integration of virtual and real environments to improve decision-making, operational efficiency, and overall healthcare delivery. By combining real-world hospital operations with virtual simulations, this approach enables better analysis, prediction, and control of complex medical systems. At the core of the parallel hospital concept lies the ACP theory, which includes artificial scenarios, computational experiments, and parallel execution. Artificial scenarios are used to model real-world hospital environments in a virtual space, allowing healthcare administrators to simulate different situations. Computational experiments are then conducted to analyze possible outcomes and optimize strategies. Finally, parallel execution ensures that insights gained from virtual simulations are applied effectively in real-world hospital operations. This continuous interaction between virtual and real systems helps in improving resource management, patient care, and emergency response. This article aims to propose a comprehensive framework for building a parallel hospital system. To validate the practicality and scientific reliability of this concept, a case study of Tiantan Hospital's "Tiantan Smart Brain" is presented. The case study demonstrates how virtual-real interaction can enhance hospital management, thereby confirming the feasibility of implementing parallel hospital systems in modern healthcare environments.

LINK: [Parallel Hospital: ACP-Based Hospital Smart Operating System | IEEE Conference Publication | IEEE Xplore](#)

PAPER-3:

TITLE: HospiT'Win: A Predictive Simulation-Based Digital Twin for Patients Pathways in Hospital

AUTHORS: Abdallah Karakra; Franck Fontanili; Elyes Lamine; Jacques Lamothe

YEAR: 2019

INTRODUCTION:

Modern hospitals are increasingly transitioning from traditional system-based operations to advanced information technology-driven environments. This transformation is fueled by the need to enhance the quality, efficiency, and responsiveness of patient care. Emerging technologies such as the Internet of Things (IoT), Body Sensor Networks (BSN), modelling and simulation, and Artificial Intelligence (AI) are becoming essential components of next-generation healthcare systems. These technologies enable continuous data collection, real-time monitoring, and intelligent analysis of patient conditions, which are crucial for delivering personalized and timely medical care.

In this context, collecting and analysing patient data, along with monitoring their physiological states and behaviours, has become a fundamental requirement in modern healthcare. To address these needs, this paper proposes a novel framework called HospiT'Win, designed to support the concept of the hospital of the future. The framework leverages advanced technologies to create a digital twin, which is a virtual replica of the hospital environment. This digital twin allows healthcare providers to track patient pathways, monitor their behaviour, and predict future health outcomes. By synchronizing real-world hospital operations with the virtual model, HospiT'Win enables more accurate decision-making and proactive healthcare management. In the event of unexpected situations, the system can simulate potential impacts and suggest effective solutions to minimize disruptions. Ultimately, this approach ensures that patients receive the right care, at the right place, and at the right time, thereby maintaining high standards of healthcare quality.

LINK: [HospiT'Win: A Predictive Simulation-Based Digital Twin for Patients Pathways in Hospital | IEEE Conference Publication | IEEE Xplore](#)

PAPER-4:

TITLE: Estimating Potential Infection Transmission Routes in Hospital Wards Using Wearable Proximity Sensors

AUTHORS: Philippe Vanhems, Alain Barrat; Ciro Cattuto; Jean-François Pinton; Nagham Khanafer; Corinne Régis; Byeul-a Kim; Brigitte Comte; Nicolas Voirin

YEAR: 2023

INTRODUCTION:

Contacts among patients, between patients and healthcare workers (HCWs), and among HCWs are a major route for the transmission of hospital-acquired infections (HAIs), making the detailed characterization of contact patterns essential for understanding HAI epidemiology and designing effective control measures. In this study, wearable sensors were used to detect close-range interactions within a geriatric unit of a university hospital, capturing contacts with a spatial resolution of approximately 1.5 meters and a temporal resolution of 20 seconds over a period of four days and four nights involving 46 HCWs and 29 patients. A total of 14,037 contacts were recorded, with 94.1% occurring during daytime, and significant variations observed in the number and duration of contacts across mornings, afternoons, and nights. Contact matrices revealed consistent mixing patterns between HCWs and patients across days, despite individual-level heterogeneity. Notably, 38% of all contacts occurred between HCWs, and a small subset of six HCWs accounted for 42% of all contacts involving at least one patient, indicating the presence of potential super-spreaders. These findings demonstrate that wearable sensors are a powerful and novel tool for quantifying hospital contact patterns, revealing both strong variability in individual contact behavior and remarkable statistical stability in overall mixing structures across days. Such high-resolution contact data are crucial for accurately informing mathematical models of HAIs and for guiding the design, evaluation, and optimization of targeted infection prevention and control strategies.

LINK: [Estimating Potential Infection Transmission Routes in Hospital Wards Using Wearable Proximity Sensors | PLOS One](#)

CHAPTER-III

METHODOLOGY

3.1 EXISTING SYSTEM

As per the information obtained from hospitals and healthcare systems, we understood the existing working system for infection control and surveillance. Multi-Drug Resistant (MDR) infections are increasing rapidly, but most hospitals still rely on traditional and manual methods to monitor and control infection spread. These systems depend heavily on delayed laboratory reports, manual contact tracing, and fragmented data sources. In the current system, infection detection is not real-time. Hospitals identify outbreaks only after laboratory confirmation, which may take several days or even weeks. During this delay, infections can spread across patients, healthcare workers, and hospital wards without immediate control. The existing system mainly involves the following processes:

- Manual data entry of patient records
- Laboratory testing and delayed reporting
- Manual contact tracing of infected patients
- Separate storage of EHR, lab, and staff data (fragmented systems)
- Reactive response after infection detection

ADVANTAGES

- The hospital system provides identification details to patients and records their medical history. This helps the hospital in future diagnosis and treatment. Hospital systems may use patient ID, Aadhaar, or hospital ID as a unique identification number.
- Hospitals can analyse previous infection cases and patient history for reference.
- Healthcare centres can generate reports and maintain documentation for infection control.
- The hospital system enables doctors to access patient records and laboratory results from centralized databases.
- A centralized hospital information system helps staff to track patient details and identify possible infection cases if required data is available. Hence, basic infection monitoring is possible.

DISADVANTAGES

- Manual contact tracing is time-consuming and inefficient
- Delay in laboratory reports leads to late outbreak detection
- Fragmented data systems make it difficult to track infection spread
- Lack of real-time monitoring increases risk of infection transmission

3.1. PROPOSED SYSTEM

The hospital has to first implement the MDR Sentinel system. The healthcare staff and patients will be provided with BLE tags or badges for real-time tracking. The system continuously collects and processes data from multiple sources. Hospital administrators and infection control teams can access the system through a dashboard interface. The system integrates patient data, lab reports, and movement tracking into a unified platform. The user (hospital staff/admin) is allowed to monitor and analyze infection risk in real time. Once the system is active, it provides various functionalities for infection control. The user will get various options on screen:

- Real-time tracking of patients and staff
- Infection risk prediction
- Contact tracing and exposure detection
- Hotspot identification
- Alert notifications
- Digital twin visualization
- Outbreak replay and analysis

3.2.SYSTEM ARCHITECTURE:

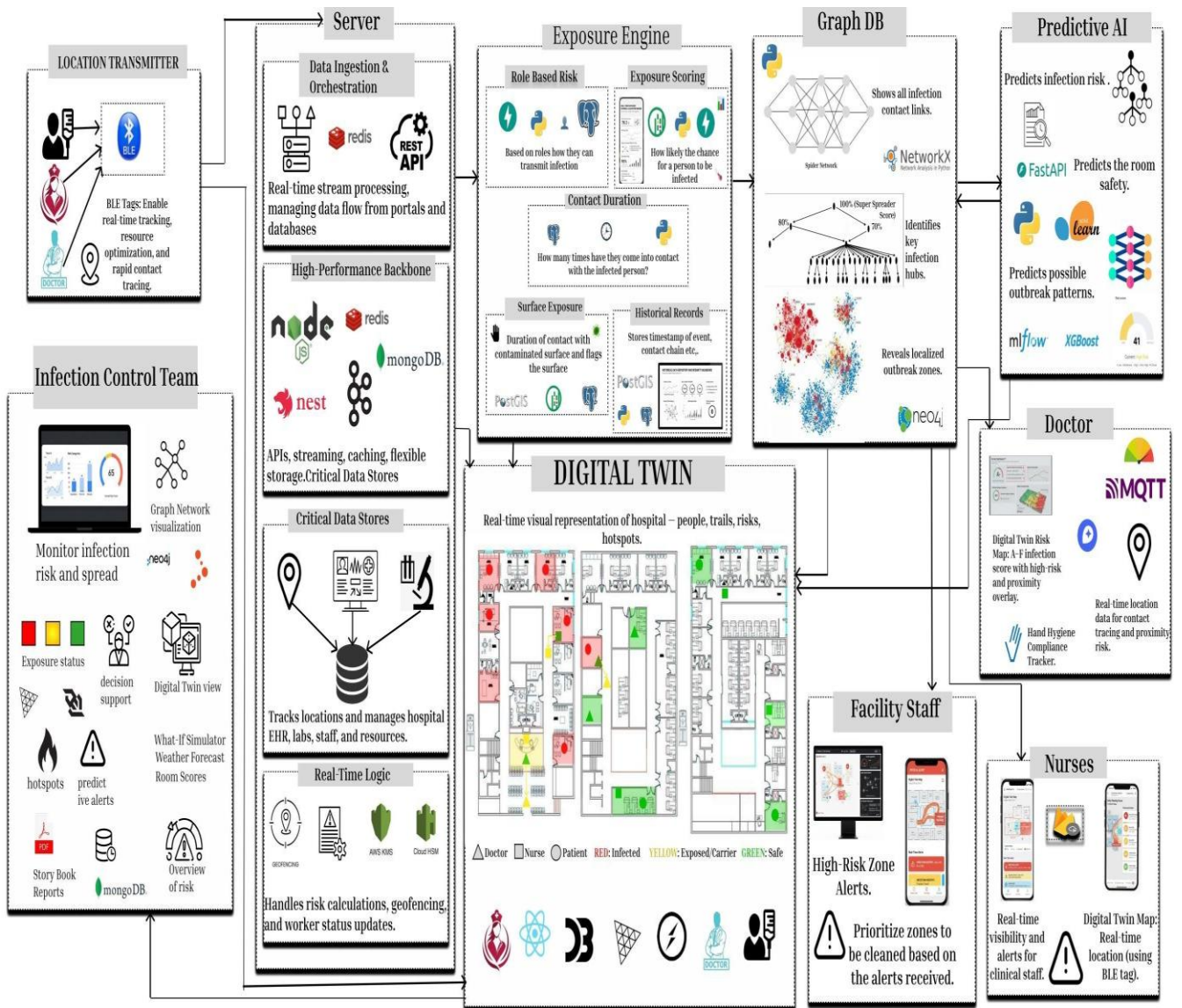


Figure-3.3.1

The user (hospital staff or administrator) can select any of the available options on the dashboard, and according to the selected option, he/she will get the required information. The user can also view real-time infection risk, high-risk contacts, and potential outbreak zones. The movement paths of patients and staff within the hospital are tracked using BLE tags and indoor mapping to monitor exposure accurately. All patient, staff, and environmental data are saved in a secure database, and only authorized administrators have access (*Refer Fig. 3.3.1*). Private and confidential information of patients and staff can only be viewed by designated personnel. This system minimizes manual paperwork and allows hospital infection control teams to monitor, analyze, and respond to MDR infection risks efficiently. With MDR Sentinel, users do not have to rely on delayed lab reports or manual contact tracing; they can access real-time information directly from the system.

APPLICATIONS:

- Real-time tracking of patients, doctors, and staff
- Identification of high-risk contacts and exposure events
- Prediction of MDR infection outbreaks before they occur
- Digital twin visualization of hospital wards for quick decision-making
- Automated alerts to infection control teams for immediate action
- Replay of infection spread to analyse root causes
- Integration of EHR, lab, and sensor data into a centralized system
- Minimization of errors in infection monitoring
- Support for proactive infection prevention and hospital-wide safety
- Continuous monitoring of critical areas to reduce hospital-acquired infection rates
- Improved compliance and operational efficiency through data-driven dashboards

3.4 SYSTEM CONFIGURATION:

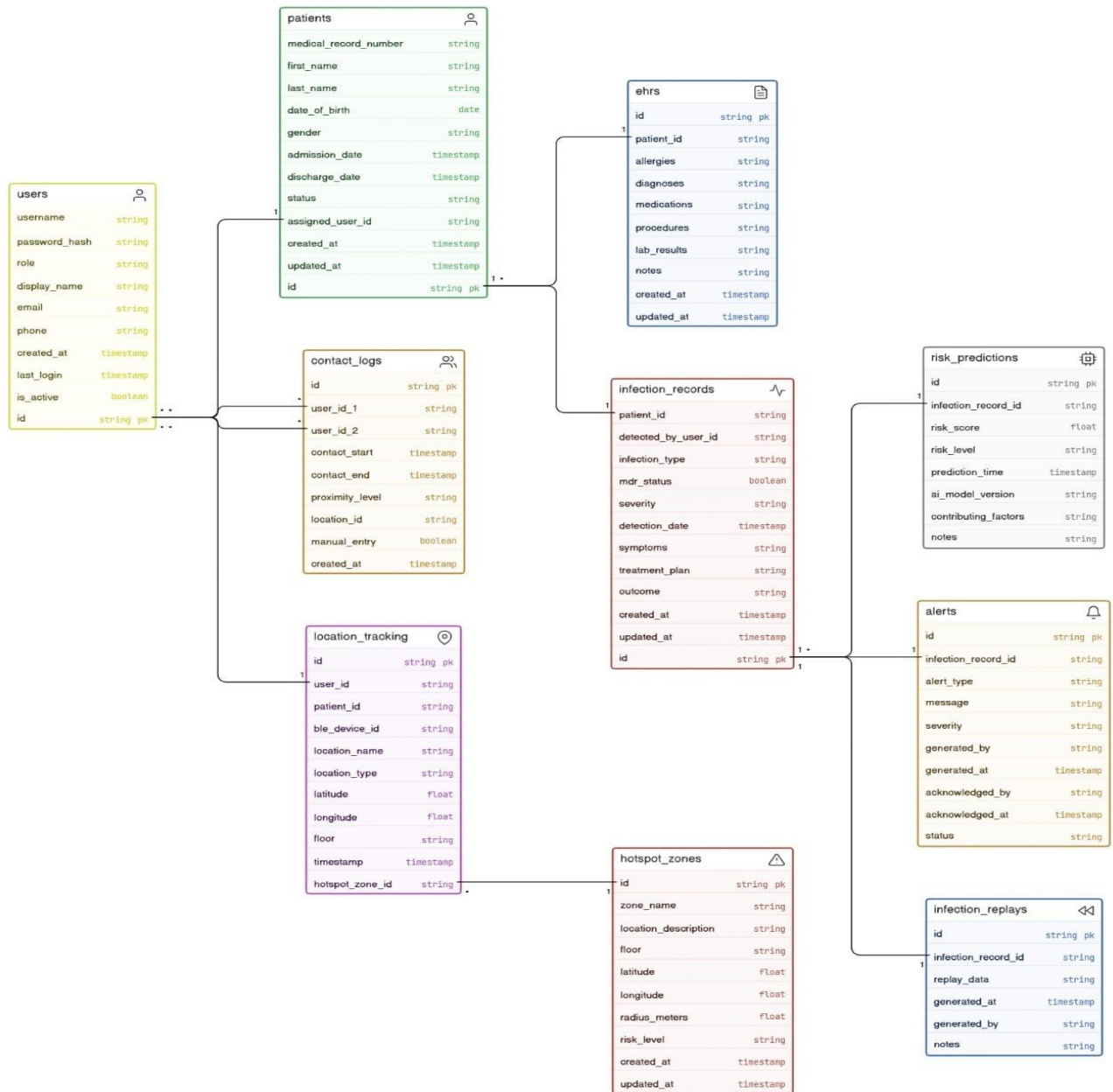
HARDWARE REQUIREMENTS:

- BLE Tags / Badges (for patients and staff tracking)
- BLE Receivers / Sensors (installed across hospital zones)
- Wi-Fi Router / Network Infrastructure
- Server System (Local Server / Cloud Access Device)
- Computers / Monitoring Systems (for dashboard access)
- Mobile Devices (for alerts and notifications)
- Power Supply Units

SOFTWARE REQUIREMENTS:

- **Operating System:** Linux (Ubuntu 22.04 LTS) or Windows
- **Frontend:** HTML, CSS, JavaScript, React.js
- **Backend:** Node.js (Express.js)
- **Database:** Neo4j (Graph Database), PostgreSQL / MongoDB
- **AI/ML Models:** Python (Scikit-learn, TensorFlow / PyTorch)
- **IoT Communication:** MQTT / WebSockets
- **Visualization:** Digital Twin Dashboard (2D/3D Web-based Interface)
- **APIs & Integration:** FHIR APIs / Hospital EHR Integration
- **Cloud Platform:** AWS / Firebase / Azure

3.5.1 ER DIAGRAM



ER Diagram Description — MDR SENTINEL (*Refer Fig 3.5.1*)

1. Core Entity: User

The User entity is the central component of the system, representing all individuals interacting with the platform.

Attributes:

- UserID
- Name
- Role (Doctor, Nurse, Patient, Admin, Infection Control Team)
- Contact Details
- Department / Ward
- Login Credentials

Users interact with patient data, alerts, infection tracking, and monitoring systems.

2. Patient & Health Data Entities

Patient

Stores patient-specific medical and hospital information:

- PatientID
- Name, Age, Gender
- Admission Date
- Diagnosis
- Ward/Room

Electronic Health Record (EHR)

Maintains medical history:

- RecordID
- PatientID (FK)
- Diagnosis History
- Medications
- Lab Reports

3. Infection Monitoring Entities

Infection Record

Captures infection-related details:

- InfectionID
- PatientID (FK)
- Infection Type (MDR)
- Detection Date
- Severity Level

Contact Log

Tracks interactions between individuals:

- ContactID
- Person1ID, Person2ID
- Contact Duration
- Timestamp
- Location

Location Tracking (BLE Data)

Captures real-time movement:

- LocationID
- UserID
- Timestamp
- Zone/Ward

4. Analysis & Prediction Entities

Risk Prediction

Generated using AI models:

- PredictionID
- PatientID/UserID
- Risk Score
- Prediction Date
- Risk Level

Hotspot Zone

Identifies high-risk areas:

- ZoneID
- Location Name
- Risk Level
- Active Status

5. Alert & Response Entities

Alert System

Generates real-time alerts:

- AlertID
- Type (High Risk / Exposure)
- Timestamp
- Status
- Triggered By (System/User)

Infection Replay

Stores past infection spread data:

- ReplayID
- InfectionID
- Timeline Data
- Visualization Link

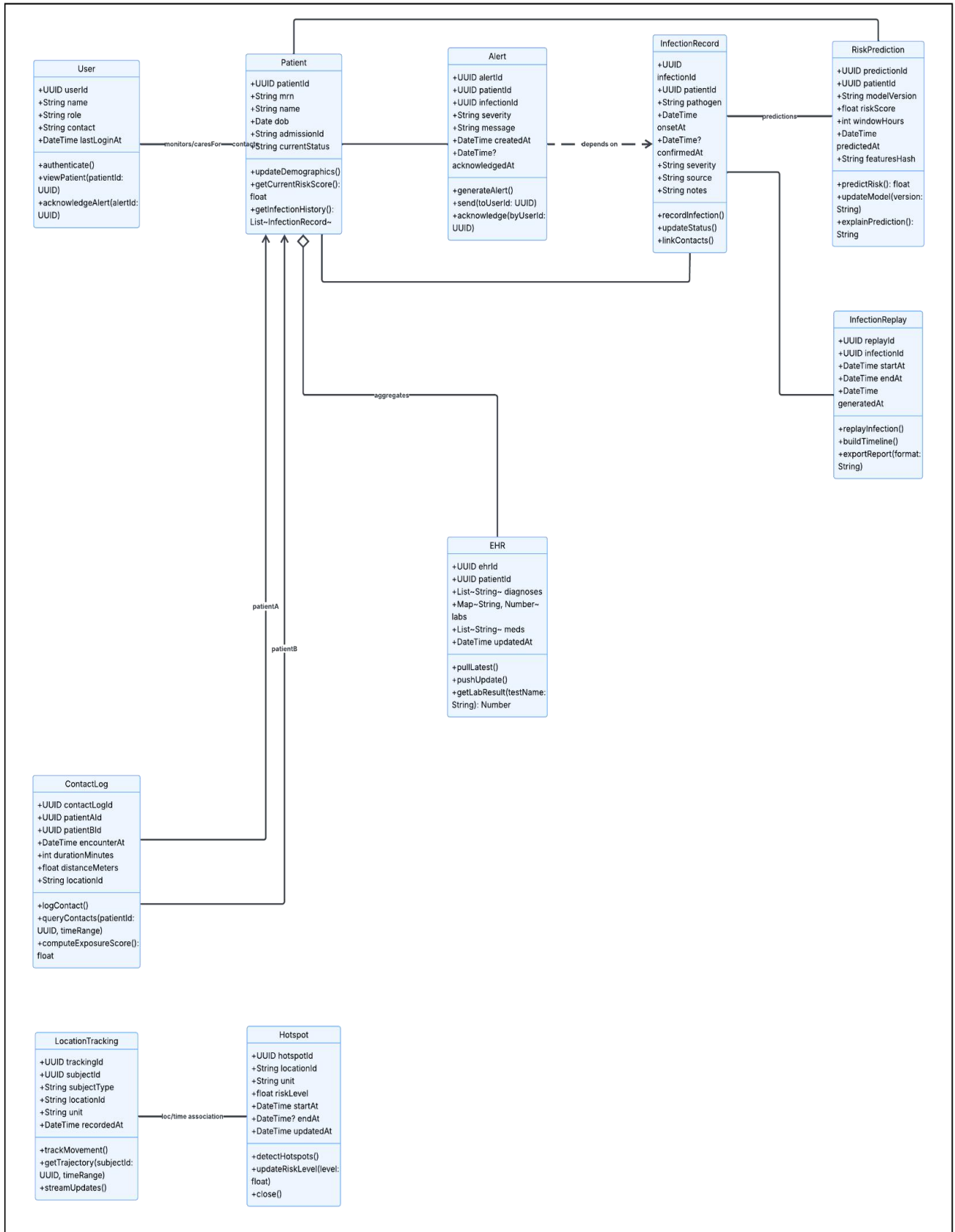
6. Relationships and Data Flow

Relationship	Type
User — Patient	One-to-Many
Patient — EHR	One-to-One
Patient — Infection Record	One-to-Many
User — Contact Log	Many-to-Many
User — Location Tracking	One-to-Many
Infection Record — Risk Prediction	One-to-One
Location Tracking — Hotspot Zone	Many-to-One
Infection Record — Alert	One-to-Many
Infection Record — Infection Replay	One-to-One

7. Purpose of Data Model

- Enables real-time infection tracking
- Supports AI-based outbreak prediction
- Improves contact tracing accuracy
- Helps identify high-risk zones (hotspots)
- Provides data-driven decision support

3.5.2 CLASS DIAGRAM



Class Diagram Description for MDR SENTINEL (*Refer Fig 3.5.2*)

- The Class Diagram of the MDR Infection Intelligence System represents the main components of the application and how they interact to support real-time infection monitoring and outbreak prevention. The central class in the system is User, which includes attributes such as UserID, Name, Role, Contact Details, and Department. Users such as doctors, nurses, administrators, and infection control staff interact with the system to manage patient data, monitor infections, and respond to alerts.
- Patient management is handled by the Patient class, which contains details such as PatientID, Name, Age, Diagnosis, and Admission Date. Each patient is associated with the EHR (Electronic Health Record) class, which stores medical history, lab reports, and treatment details. Infection-related information is managed by the InfectionRecord class, which includes attributes like InfectionID, Infection Type, Severity, and Detection Date.
- The system tracks interactions using the ContactLog class, which records contact details such as participants, duration, and timestamp. Real-time movement of patients and staff is handled by the LocationTracking class using BLE technology, storing data like LocationID, UserID, Timestamp, and Zone.
- Advanced analysis is supported by the RiskPrediction class, which uses AI models to generate risk scores and predict potential infection spread. The Hotspot class identifies high-risk areas within the hospital based on movement and infection data.
- The Alert class is responsible for generating real-time notifications related to infection risks, exposures, and hotspots, enabling quick response from healthcare staff. Additionally, the InfectionReplay class allows retrospective analysis by storing and visualizing past infection spread patterns.
- Overall, the system integrates multiple classes to provide a smart, data-driven approach for infection monitoring, helping hospitals improve patient safety, reduce infection spread, and enhance decision-making efficiency.

IV. RESULTS:

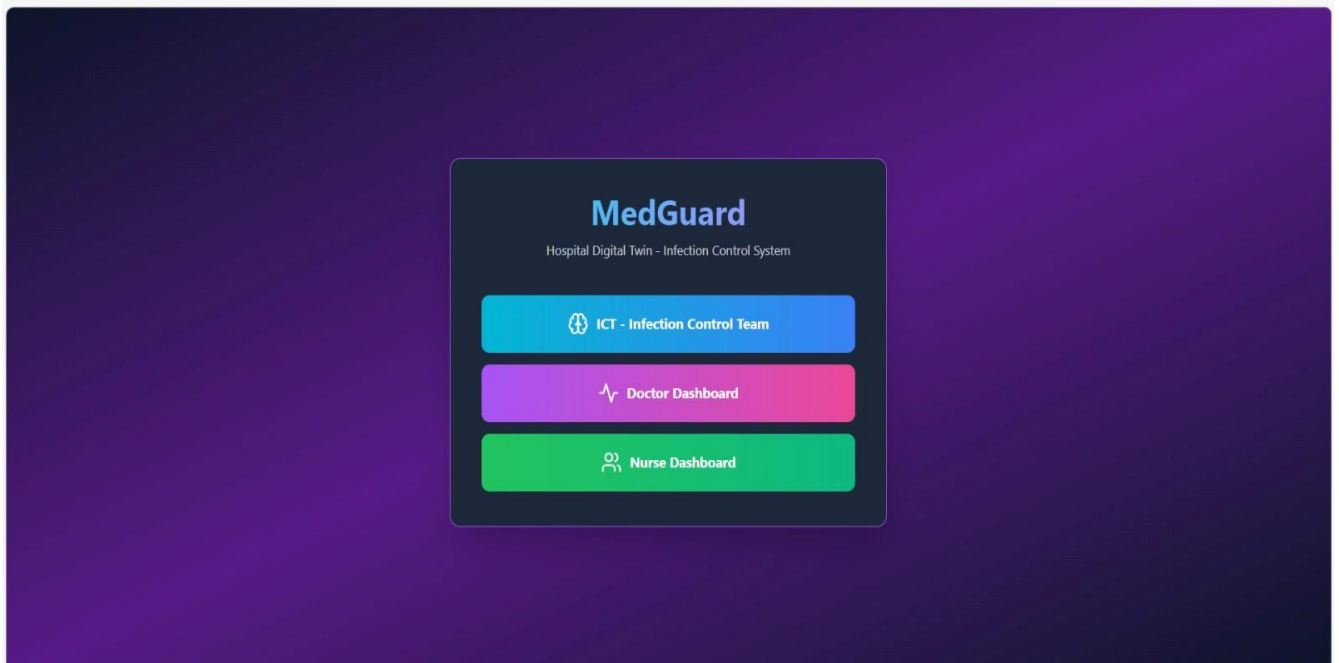


Fig-4.1

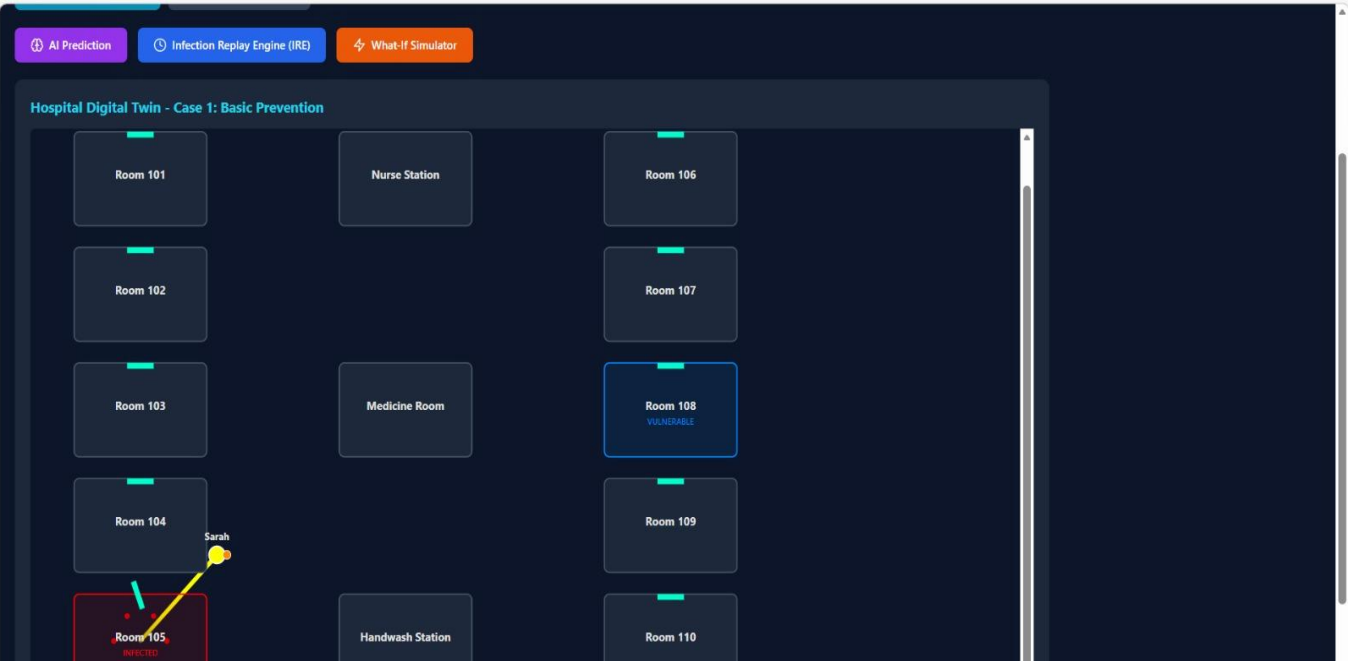
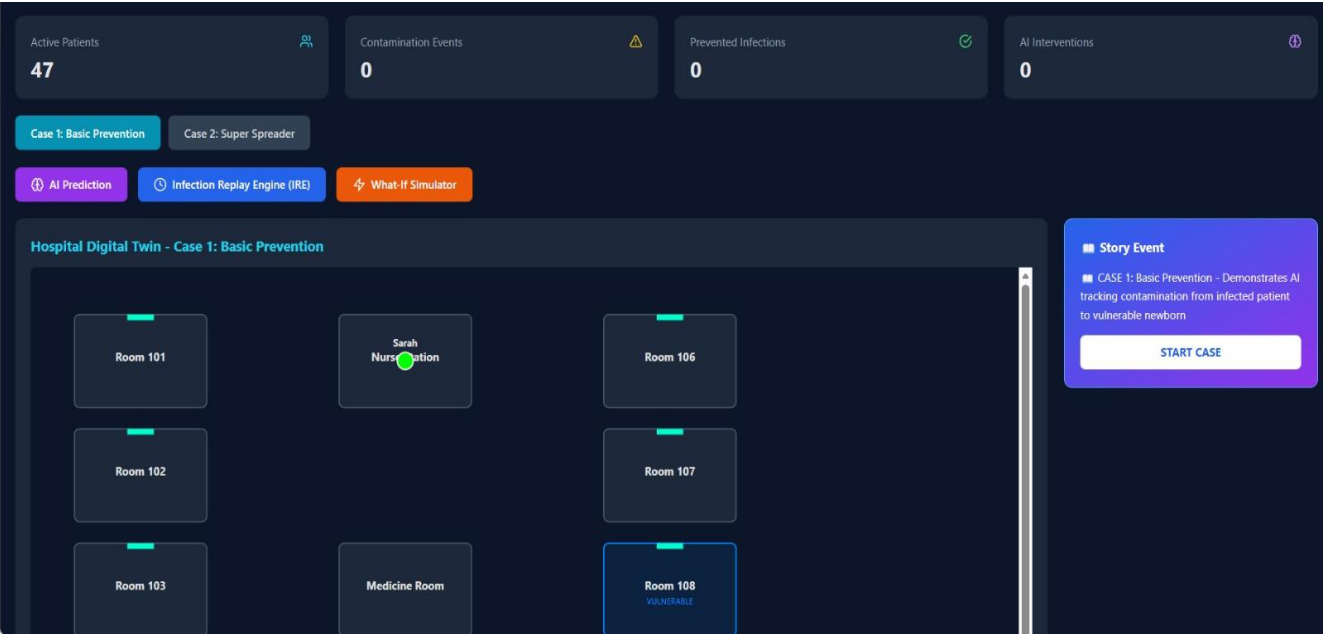


Fig-4.2

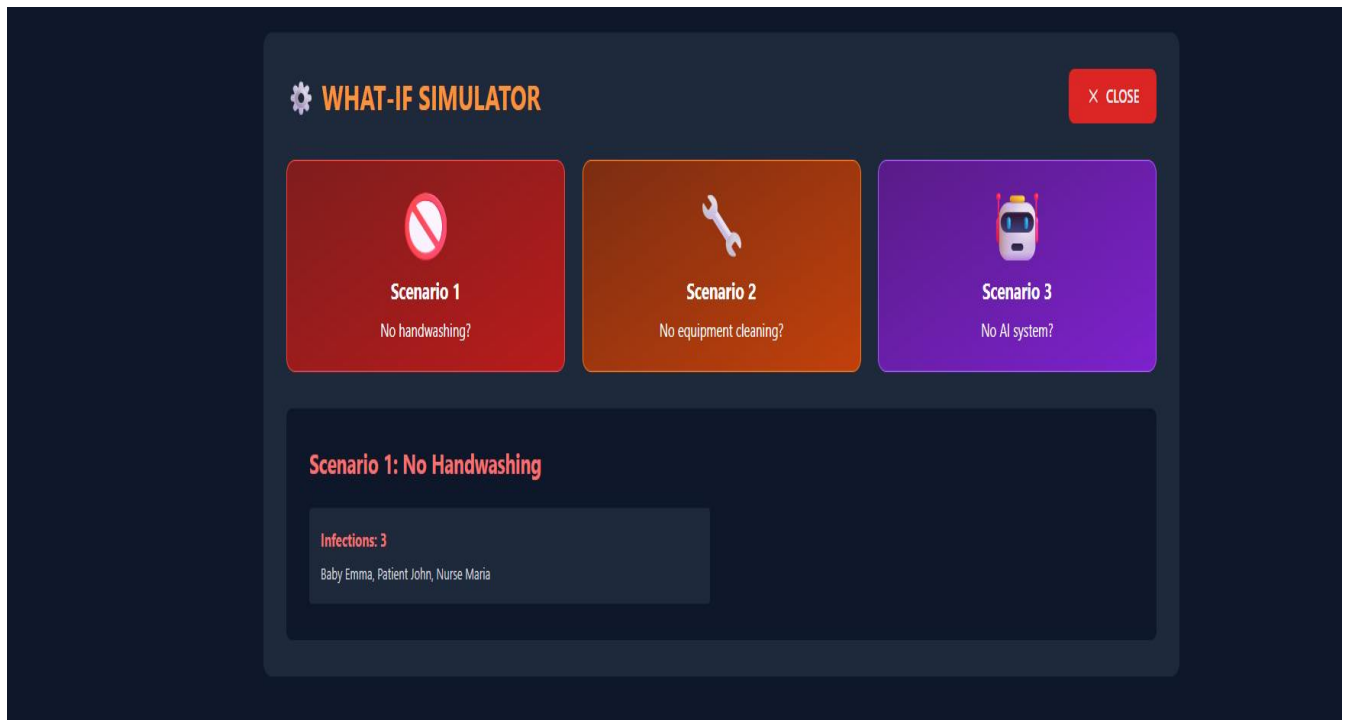
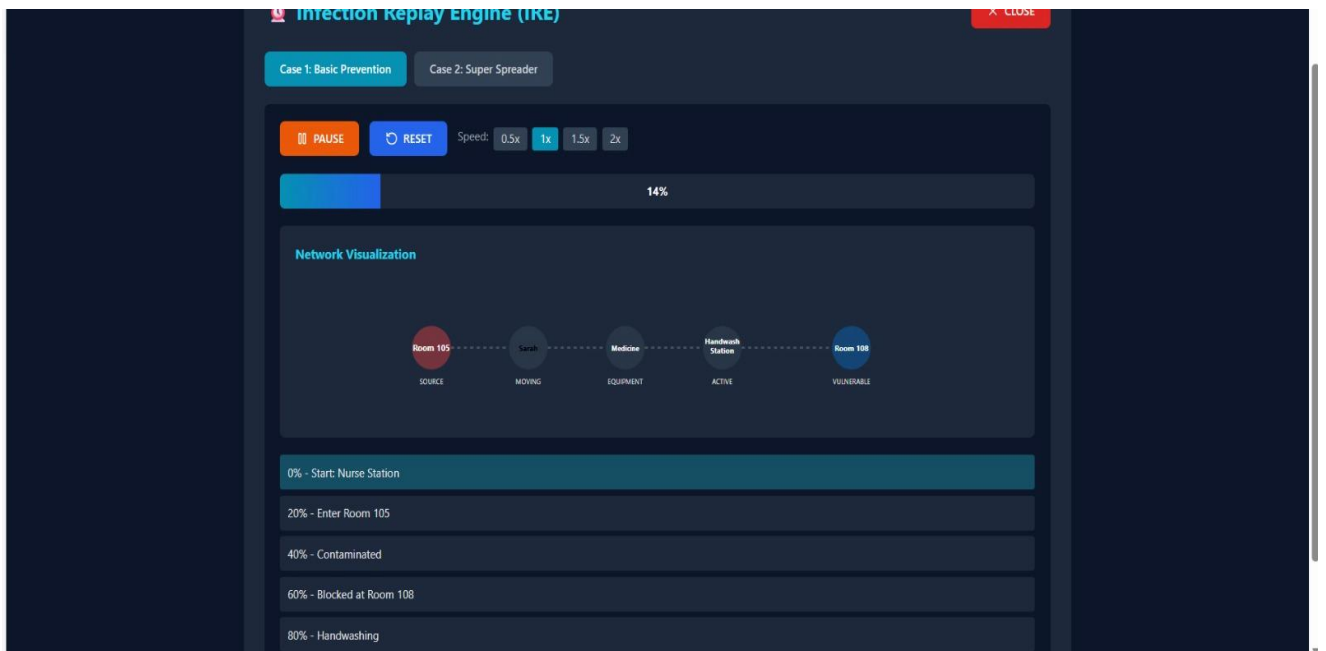


Fig-4.3

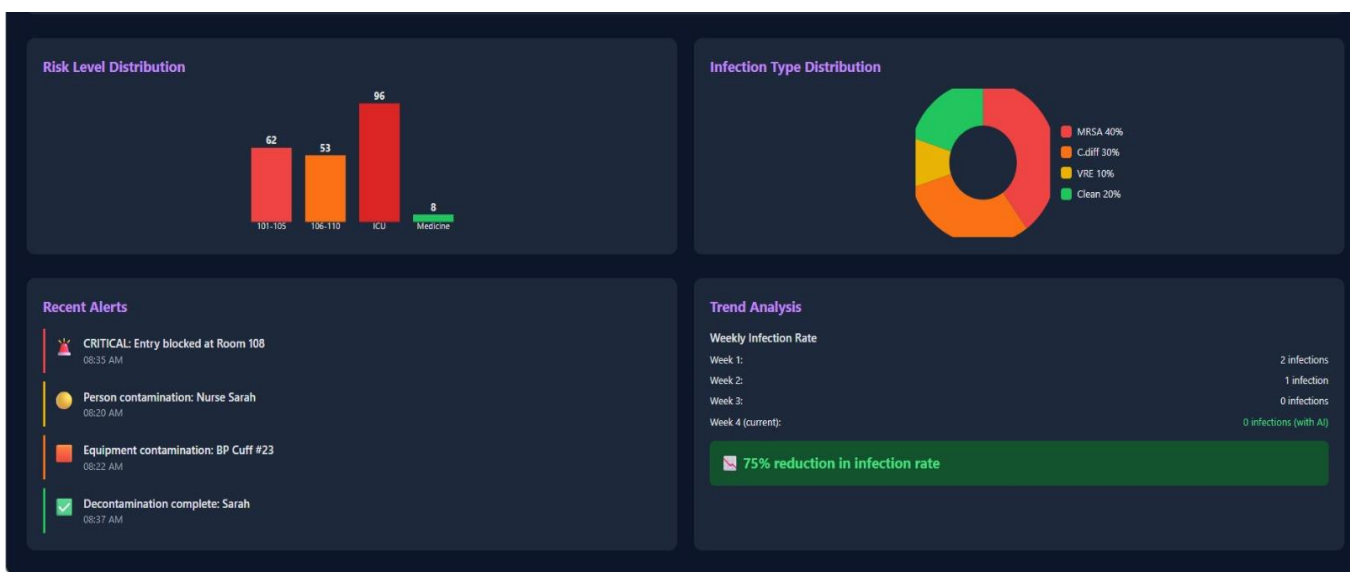
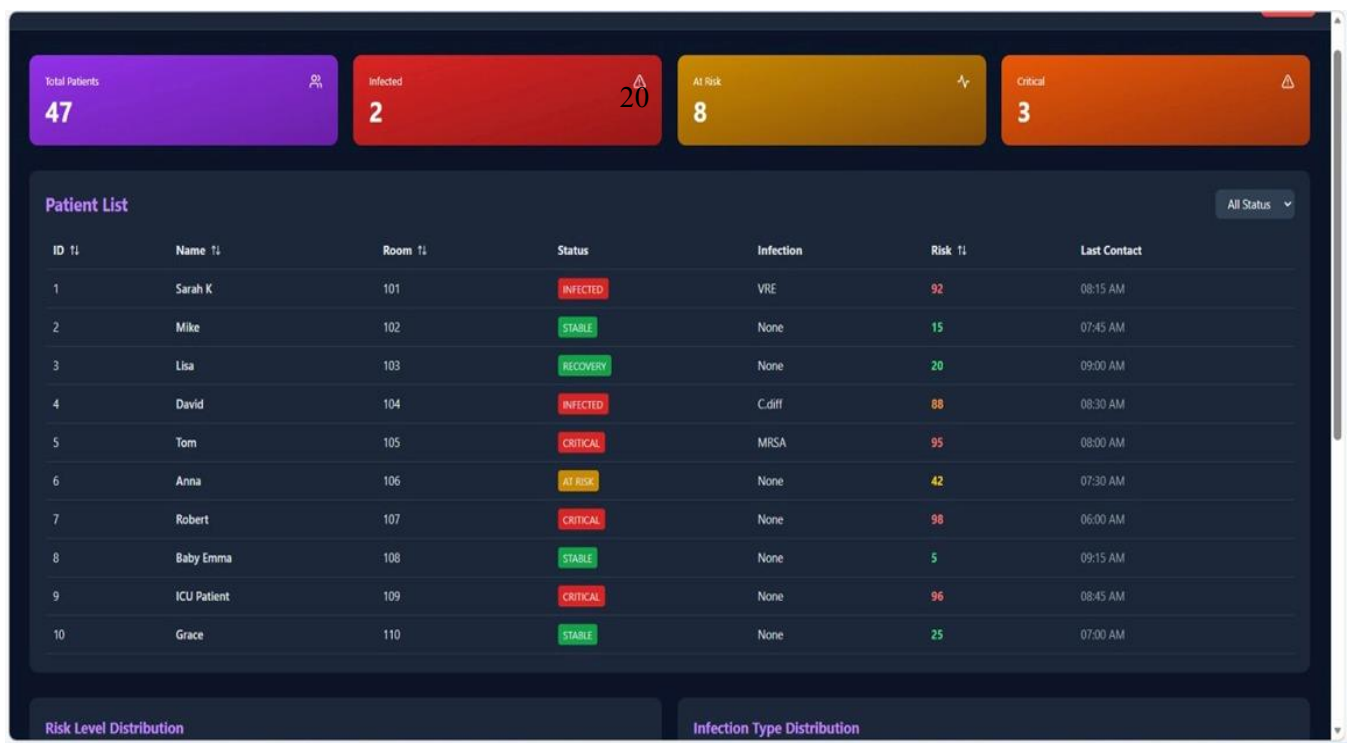


Fig – 4.4

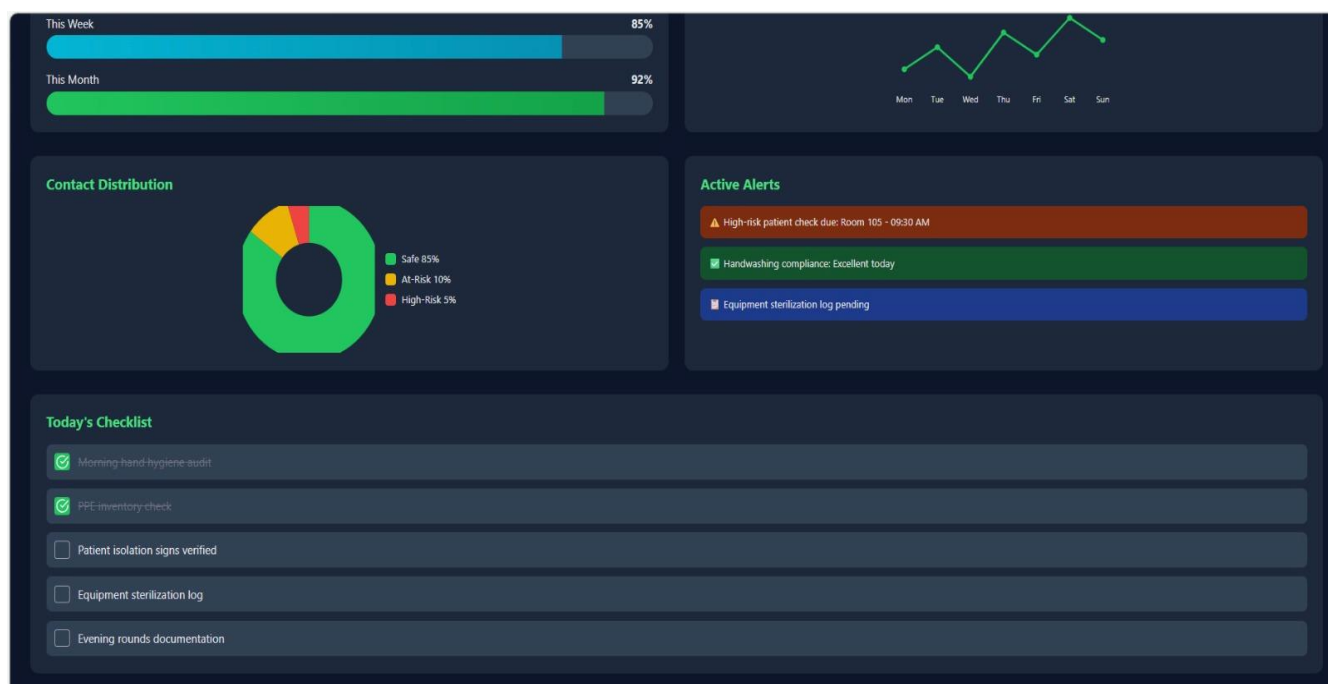
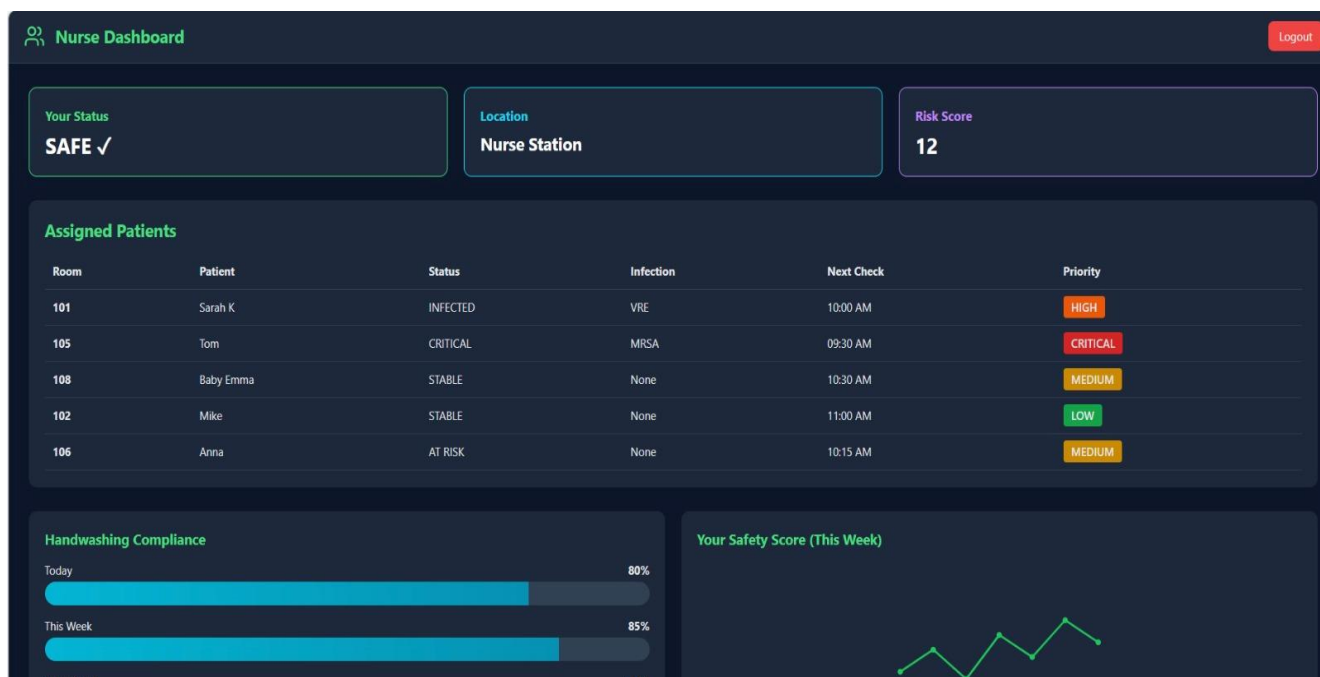


Fig – 4.5

CONCLUSION:

The proposed system, MDR Sentinel: Hospital Outbreak Intelligence, provides an effective solution for the early detection and prevention of Multi-Drug Resistant (MDR) infections in hospital environments. The existing system relies on manual surveillance, delayed laboratory reports, and fragmented data, which leads to late identification of outbreaks and increased risk of infection spread. MDR Sentinel overcomes these challenges by introducing an AI-driven, data-driven, and real-time surveillance system. By integrating Bluetooth Low Energy (BLE) technology, digital twin modelling (*Refer Fig. 4.2*), and machine learning algorithms, the system enables continuous monitoring of patients, healthcare staff, and their interactions. This helps in identifying high-risk contacts, detecting infection hotspots, and predicting potential outbreaks before they occur. The system also provides real-time alerts, visualization dashboards, and outbreak replay features, which support faster decision-making and effective infection control (*Refer Fig. 4.3*). By transforming the traditional reactive approach into a proactive and intelligent system, MDR Sentinel improves patient safety, reduces infection rates, and enhances overall hospital efficiency. In conclusion, MDR Sentinel represents a significant step toward smart healthcare systems by enabling accurate, timely, and data-driven infection management. Future enhancements can further improve its performance by integrating advanced IoT sensors and intelligent treatment recommendations, making it a powerful tool for modern hospitals.

REFERENCES

- [1] Lindsay James., Digital twins will revolutionise healthcare: Digital twin technology has the potential to transform healthcare in a variety of ways – improving the diagnosis and treatment of patients, streamlining preventative care and facilitating new approaches for hospital planning., 2021
- [2] Meghan A Baker, Deborah S Yokoe , John Stelling, Ken Kleinman, Rebecca E Kaganov, Alyssa R Letourneau, Neha Varma, Thomas O'Brien, Martin Kulldorff, Damilola Babalola, Craig Barrett, Marci Drees, Micaela H Coady, Amanda Isaacs, Richard Platt, Susan S Huang., Automated outbreak detection of hospital-associated pathogens: Value to infection prevention programs., 2020
- [3] Andrea Madotto, Jiming Liu., Super-Spreader Identification Using Meta-Centrality., 2016
- [4] Johannes Müller Volker Hösel., Available: [Contact tracing & super-spreaders in the branching-process model - PMC.](#), 2023
- [5] Philippe Vanhems,Alain Barrat,Ciro Cattuto, Jean-François Pinton, Nagham Khanafer, Corinne Régis, Byeul-a Kim,Brigitte Comte,Nicolas Voirin., Available: [Estimating Potential Infection Transmission Routes in Hospital Wards Using Wearable Proximity Sensors | PLOS One.](#), 2023
- [6] Michelle Lin., Combating Multidrug-Resistant Gram-Negative Bacteria Using Biofilm Protein Pase 1 as Novel Dispersal Agent for Co-Treatment Therapy., 2023
- [7] Abdallah Karakra; Franck Fontanili; Elyes Lamine; Jacques Lamothe., Available: [HospiT'Win: A Predictive Simulation-Based Digital Twin for Patients Pathways in Hospital | IEEE Conference Publication | IEEE Xplore.](#), 2019
- [8] Lindsay James., Design and Integration of IoRT-Enabled Robotic Systems with Wearable Devices in a Digital Twin Framework., 2024
- [9] Yongjun Wang; Fei-Yue Wang; Xiao Wang; Zhongjun Guan; Liwei Ouyang; Jiangong Wang., Parallel Hospital: Acp-Based Hospital Smart Operating System., 2021
- [10] Xiaohua Hu, Hongwei Cao, Jia Sh, Yonghui Dai, Weihui Dai., Study of Hospital Emergency Resource Scheduling Based on Digital Twin Technology., 2022